

# Dheeran E. Wiggins

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## EDUCATION

IN PROGRESS **BS Mathematics**, *University of Illinois Urbana-Champaign*  
JUNE 2023 **HS Diploma**, *Illinois Mathematics and Science Academy*

## RESEARCH EXPERIENCE

JUN 2026 – PRESENT **Researcher**, *Institute for Quantum Computing, University of Waterloo*  
in Quantum Channels. Advised by Graeme Smith.

DEC 2025 – PRESENT **Undergraduate Researcher**, *Department of Mathematics, University of Illinois*  
in Defect TQFT. Advised by James Pascaleff.

JAN 2024 – JUN 2025 **Research Scholar**, *Illinois Mathematics Lab, University of Illinois*  
in Quantum Graphs. Advised by Roy Araiza.

JUN 2025 – JUL 2025 **Fellow**, *MathQuantum Research Training Group, University of Maryland*  
in Ultraproducts and Density Operators. Advised by Maicol Ochoa.

APR 2022 – SEP 2024 **Undergraduate Research Fellow**, *Endeavor Health Research Institute*  
in Mathematical Epidemiology and Dynamics. Advised by Anthony Solomonides.

JUN 2023 – AUG 2023 **Research Assistant**, *Northwestern University*  
in Biomedical Engineering. Advised by Jules Dewald and Hongchul Sohn.

JUN 2022 – MAR 2023 **Research Collaborator**, *University of Chicago*  
in COVID-19 Outcomes Data Analysis. Under Julian Solway.

MAY 2021 – JUN 2023 **Student Researcher**, *Fermilab*  
in Dark Photon and Doubly Charged Higgs Searches. Advised by Peter Dong.

## RELEVANT COURSEWORK

### University of Illinois Urbana-Champaign

$\mathcal{G}$  denotes graduate.

- Geometric Group Theory $\mathcal{G}$
- Group Representation Theory $\mathcal{G}$
- Topological Field Theories in 2D $\mathcal{G}$
- Rep-theoretic Methods in QIT $\mathcal{G}$
- Abstract Algebra I, II $\mathcal{G}$
- Intro to Abstract Algebra II
- Homotopy Theory $\mathcal{G}$
- Algebraic Topology I, II $\mathcal{G}$
- General Topology $\mathcal{G}$
- Quantum Info Processing Theory
- Advanced Symbolic Logic
- $\mathcal{H}$  denotes honors.
- Abstract Linear Algebra $\mathcal{H}$
- Honors Real Analysis $\mathcal{H}$
- Differential Equations
- Symbolic Logic
- Fundamental Mathematics $\mathcal{H}$
- Probability Theory

### Illinois Mathematics and Science Academy

$\mathcal{I}$  denotes intersession.

- Category Theory $\mathcal{I}$
- Abstract Algebra
- Number Theory
- Linear Algebra
- Multi-variable Calculus
- Single Variable Calculus
- Quantum Mechanics
- Intro to Quantum Computing $\mathcal{I}$
- Modern Physics
- Sound & Light
- Electricity & Magnetism
- Classical Mechanics

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## TALKS

- MAY 2026 **Some (semi)simple things about 2D unitary TQFT with defects**, *MathQuantum Annual Symposium, University of Maryland*, Virtual
- APR 2026 **Unitary field theories with defects in two dimensions**, *Illinois Undergraduate Research Symposium, University of Illinois*, Champaign, IL  
Talk advised by James Pascaleff.
- MAR 2026 **Locality via (co)limits in an infinite spin chain**, *Rose-Hulman Undergraduate Mathematics Conference, Rose-Hulman Institute of Technology*, Terre-Haute, IN  
with Igor Mineyev.
- MAR 2026 **Stabilizer formalism in  $\infty$ -qubit systems**, *APS Global Physics Summit*, Denver, CO  
with Igor Mineyev.
- APR 2025 **A model for the infinite tensor product of groups: Insights from quantum information**, *Illinois Undergraduate Research Symposium, University of Illinois*, Champaign, IL  
Talk advised by Igor Mineyev.
- MAR 2025 **On noncommutative graphs and Poulin's stabilizer formalism**, *Rose-Hulman Undergraduate Mathematics Conference, Rose-Hulman Institute of Technology*, Terre-Haute, IN  
Talk advised by Roy Araiza.
- MAY 2024 **Winter Spaces and the Stabilizer Formalism**, *Illinois Mathematics Lab Open House, University of Illinois*, Champaign, IL  
with Anderson et al. Poster advised by Roy Araiza.
- APR 2024 **Modeling a Viral Epidemic With a Concurrent "Misinfodemic"**, *Research Institute Spring Scientific Research Poster Reception*, Evanston, IL  
Poster advised by Anthony Solomonides.
- APR 2024 **Modeling a Viral Epidemic With a Concurrent "Misinfodemic"**, *Illinois Undergraduate Research Symposium, University of Illinois*, Champaign, IL  
Talk advised by Anthony Solomonides.
- APR 2024 **Winter Spaces and the Stabilizer Formalism**, *Illinois Quantum Information Science & Technology (IQUIST) World Quantum Day, University of Illinois*, Champaign, IL  
with Anderson et al. Poster advised by Roy Araiza.
- APR 2023 **Identification and estimated yield of background events in a search for a doubly charged Higgs boson at CMS**, *APS April Meeting*, Minneapolis, MN  
with Anne et al. Poster advised by Peter Dong.
- APR 2023 **Identification of optimal production channels for dark photon searches**, *APS April Meeting*, Minneapolis, MN  
with Anne et al. Poster advised by Peter Dong.
- APR 2023 **Unidirectional build architecture: Refactoring a HEP data collection codebase**, *Annual IMSAloquium, Illinois Math and Science Academy*, Aurora, IL  
Talk advised by Peter Dong.
- APR 2022 **An investigation of triboson decays into four-lepton final states**, *Annual IMSAloquium, Illinois Math and Science Academy*, Aurora, IL  
with George Bayliss and Jesus Fileto. Talk advised by Peter Dong.

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## TEACHING

- JUL 2026 **Topological Field Theories in Quantum Computing**, *University of Maryland*  
Lecture for MathQuantum 2026 High School Fellowship
- SPR 2026 **Three Solutions to the Basel Problem**, *University of Illinois*  
Project for James Scholar Honors students in Math 231: Calculus II ([course page](#)).
- SPR 2026 **Math 231: Calculus II, Discussion ADP**, *University of Illinois*
- JAN 2026 **Monoidal Categories for Quantum Theory**, *Illinois Math and Science Academy*  
Minicourse for Intersession 2026 ([course page](#))
- JAN 2026 **Intro to Quantum Computing and Algorithms**, *Illinois Math and Science Academy*  
with Anastasia Perry. Minicourse for Intersession 2026
- FAL 2025 **Math 231: Calculus II, Discussion BDD**, *University of Illinois*
- SUM 2025 **Hidden Subgroups and Quantum Computation**, *Illinois Math and Science Academy*  
Minicourse for Summer 2025 ([course page](#))
- JUL 2025 **Quantum Channels and Error Correction**, *University of Maryland*  
Lecture for MathQuantum 2025 High School Fellowship ([slides](#))
- JAN 2025 **Intro to Mathematical Quantum Error Correction**, *Illinois Math and Science Academy*  
Minicourse for Intersession 2025 ([course page](#))

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## LEARNING

- JUN 2026 **Summer School on Homotopy Colimits**, *University of Regina*, Regina, Canada
- MAY 2026 **Atlantic TQFT Spring School**, *Dalhousie University*, Halifax, Canada
- MAY 2026 **UG School on Experimental QIP**, *Institute for Quantum Computing*, Waterloo, Canada
- SPR 2026 **Modern and classical homotopy theory**, *University of Illinois*, Champaign, IL  
Reading course with Jeremiah Heller.
- SPR 2026 **Tensor categories**, *University of Illinois*, Champaign, IL  
Reading course with Charles Rezk.
- DEC 2024 – PRESENT **Infinite tensor products in quantum**, *University of Illinois*, Champaign, IL  
Independent study with Igor Mineyev.
- FAL 2025 **Monoidal, enriched, and dagger categories**, *University of Illinois*, Champaign, IL  
Reading course with Charles Rezk.
- AUG 2025 **Summer School on Geometry**, *Universiteit Utrecht*, Utrecht, Netherlands
- SPR 2025 **Rudiments of von Neumann algebras**, *University of Illinois*, Champaign, IL  
Reading seminar with Jihong Cai and Roy Araiza.
- JUL 2023 **Summer School on Condensed Matter Physics**, *Princeton University*, Princeton, NJ

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## PROFESSIONAL AFFILIATIONS

- MAR 2024 – PRESENT **Member**, *American Mathematical Society*
- FEB 2023 – PRESENT **Member**, *American Physical Society*

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## RELEVANT SKILLS

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|-------------|-------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------|
| SYMBOLIC    | SageMath, Maple, Mathematica                                                                          | NUMERICAL | MATLAB, Scilab, GNU Octave                 |
| PROGRAMMING | Python, C++, Java                                                                                     | MARKUP    | L <sup>A</sup> T <sub>E</sub> X, HTML, CSS |
| LANGUAGES   | English <i>fluent</i> , Tamil <i>spoken</i> , Russian <i>intermediate</i> , Spanish <i>elementary</i> |           |                                            |